

Checkpoint 1 Report: Negative-Weight Single-Source Shortest Paths in Near-Linear Time

Team: 68

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1 Paper Selection Proposal

1.1 Paper Details

- **Title:** Negative-Weight Single-Source Shortest Paths in Near-Linear Time
- **Authors:** Aaron Bernstein, Danupon Nanongkai, Christian Wulff-Nilsen
- **Conference:** Communications of the ACM Vol. 68, No. 2
- **Year:** 2023 (updated)
- **Link:** <https://dl.acm.org/doi/10.1145/3631536>

1.2 Summary

The paper addresses the Single-Source Shortest Paths (SSSP) problem in directed graphs with negative edge weights, a fundamental challenge in graph algorithms. Traditional methods like Bellman-Ford have high time complexity $O(mn)$, which is impractical for large graphs. Recent approaches rely on complex optimization techniques, introducing significant overhead.

The authors present a **combinatorial algorithm** achieving near-linear runtime $O(m \log^6 n \log W)$, significantly simplifying prior methods.

Key innovations include:

- **Price functions** to transform edge weights into non-negative values.
- **Low-diameter decomposition** to partition the graph into manageable components.
- **Recursive scaling** to progressively eliminate negative weights.

1.3 Justification

The paper was selected for the following reasons:

- **Theoretical Impact:** Resolves a decades-old problem with a near-optimal solution.
- **Practical Relevance:** Applicable to logistics (e.g., routes with fuel rebates), financial arbitrage, and network optimization.
- **Simplicity:** Avoids complex continuous optimization, making it more accessible for implementation.
- **Classroom Relevance:** We have already studied both algorithms in class, and it would be interesting to work on the shortcomings of those algorithms and overcome them using a single unified algorithm.

1.4 Implementation Feasibility

Resources:

- **Theoretical Foundation:** Detailed pseudocode and proofs are provided.
- **Code Repositories:** No public code exists, but standard graph libraries (e.g., NetworkX, Boost Graph) can be adapted.
- **Datasets:** Synthetic graphs with negative weights.

Theoretical Descriptions:

- Efficient handling of probabilistic graph decomposition.
- Optimizing priority queues for the hybrid Dijkstra-Bellman-Ford.

1.5 Team Responsibilities

- **Reading and Analyzing the Paper:** Muhammad Daniyal Farooqui
- **Coding and Implementation:** Jibran Sheikh, Muhammad Daniyal Farooqui
- **Writing:** Jibran Sheikh